ORIGINAL ARTICLE

Apixaban for Extended Treatment of Provoked Venous Thromboembolism

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ABSTRACT

BACKGROUND

The appropriate duration of anticoagulation for venous thromboembolism (VTE) in patients who have a transient provoking factor (e.g., surgery, trauma, or immobility) and concomitant enduring risk factors is uncertain.

METHODS

In this single-center, double-blind, randomized trial, adults with VTE after the occurrence of a transient provoking factor who had at least one enduring risk factor and had completed at least 3 months of anticoagulation were assigned to receive oral apixaban (at a dose of 2.5 mg twice daily) or placebo for 12 months. The primary efficacy outcome was the first symptomatic recurrent VTE. The primary safety outcome was the first episode of major bleeding according to the criteria of the International Society on Thrombosis and Hemostasis.

RESULTS

A total of 600 patients underwent randomization (mean age, 59.5 years; female sex, 57.0%; non-White race, 19.2%). The trial population had a broad range of provoking factors and enduring risk factors. Symptomatic recurrent VTE occurred in 4 of the 300 patients (1.3%) in the apixaban group and in 30 of the 300 patients (10.0%) in the placebo group (hazard ratio, 0.13; 95% confidence interval [CI], 0.04 to 0.36; P<0.001). Major bleeding occurred in 1 patient in the apixaban group and none in the placebo group. Clinically relevant nonmajor bleeding was observed in 14 of 294 patients (4.8%) in the apixaban group and in 5 of 294 patients (1.7%) in the placebo group (hazard ratio, 2.68; 95% CI, 0.96 to 7.43; P=0.06). One patient in the apixaban group and 3 patients in the placebo group died, with no deaths attributed to cardiovascular or hemorrhagic causes. Nonhemorrhagic, nonfatal adverse events occurred in 6 patients (2.0%) in each group.

CONCLUSIONS

Among patients with provoked VTE and enduring risk factors, low-intensity therapy with apixaban for 12 months resulted in a lower risk of symptomatic recurrent VTE than placebo, with a low risk of major bleeding. (Funded by Bristol-Myers Squibb–Pfizer Alliance; HI-PRO ClinicalTrials.gov number, NCT04168203.)

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*A complete list of the HI-PRO trial investigators is provided in the Supplementary Appendix, available at NEJM.org.

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Venous thromboembolism (VTE) has traditionally been categorized as provoked (occurring after such transient risk factors as surgery, trauma, or immobility) or unprovoked (occurring without a readily identifiable trigger).[1] Extended-duration anticoagulation is commonly prescribed for patients with unprovoked VTE because of a high risk of recurrence (6 to 10% per year) after the discontinuation of anticoagulation.[2-8] For patients with provoked VTE who do not have cancer or severe thrombophilia, the recurrence risk is lower.[9] In such patients, guidelines typically recommend 3 months of anticoagulation.[4,6,8,10] In the commonly encountered population of patients with provoked VTE and concomitant enduring risk factors (e.g., autoimmune disorders, chronic lung disease, or obesity), the appropriate duration of anticoagulation is unclear.[1,11]

We performed the Extended-Duration Low-Intensity Apixaban to Prevent Recurrence in High-Risk Patients with Provoked Venous Thromboembolism (HI-PRO) trial to assess the efficacy and safety of oral apixaban (at a dose of 2.5 mg twice daily), as compared with placebo, for the prevention of recurrence in patients with provoked VTE and at least one enduring risk factor.

METHODS

TRIAL DESIGN AND OVERSIGHT

HI-PRO was a randomized, placebo-controlled, double-blind single-center trial conducted at Brigham and Women's Hospital in Boston.[12] The trial was conducted in accordance with all local laws and regulations and the ethical principles of the Declaration of Helsinki and the International Council for Harmonisation guidelines for Good Clinical Practice. The trial was approved by the institutional review board at Brigham and Women's Hospital. An independent data and safety monitoring board met at prespecified intervals to monitor the trial.

The trial was funded by the Bristol-Myers Squibb–Pfizer Alliance, which had no direct role in the design, execution, statistical analysis, or reporting of the trial. There were no agreements concerning confidentiality of the data between the industry sponsor and the authors. The Thrombosis Research Group at Brigham and Women's Hospital was responsible for gathering and housing the individual patient data and for performing the statistical analysis. The first and last authors designed the trial, had unrestricted access to the data, had the sole responsibility for the decision to submit the manuscript for publication, and developed the protocol (available with the full text of this article at NEJM.org). The first author wrote the first draft of the manuscript. All the authors vouch for the accuracy and completeness of the reported data and the fidelity of the trial to the protocol.

The trial investigators used a secure Web-based electronic data-capture tool (REDCap) to record trial data.[13] All the patients provided written informed consent.

TRIAL POPULATION

Patients were required to be at least 18 years of age and to have objectively confirmed VTE with at least one provoking factor, including major surgery, trauma, acute medical illness, or immobility. At the time of screening, all the patients had been receiving anticoagulation for at least 3 months and had at least one enduring risk factor for recurrence, such as a body-mass index (the weight in kilograms divided by the square of the height in meters) of at least 30, chronic lung disease, or chronic inflammatory disease. Details are provided in the Supplementary Appendix, available at NEJM.org.

The presence of at least one provoking factor was identified by the patient's longitudinal clinician, and investigators confirmed the presence at enrollment. Patients could have deep-vein thrombosis in a lower or an upper limb,[14] pulmonary embolism in any location (including proximal or isolated subsegmental pulmonary arteries),[15] or both. Atherosclerotic cardiovascular disease has been recognized as a risk factor for incident[16] and recurrent VTE.[17] A diagnosis of atherosclerotic cardiovascular disease satisfied the requirement for an enduring risk factor but was restricted to no more than 35% of the patients in each trial group to control for baseline aspirin use.[16] Eligible patients could have multiple enduring risk factors consistent with data from patients with provoked VTE.[1,11,18]

Key exclusion criteria were an indication for anticoagulation (including VTE without provoking factors or associated with active cancer), a contraindication for anticoagulation, a life ex-

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pectancy of less than 12 months, pregnancy or lactation, severe hepatobiliary or renal disease, receipt of a $P2Y_{12}$ platelet receptor antagonist or an aspirin dose of more than 81 mg daily, and recent or active bleeding. Details regarding inclusion and exclusion criteiria are provided in the Supplementary Appendix.

RANDOMIZATION AND TRIAL INTERVENTION

We performed randomization using a permuted block size of 4, with 1:1 assignment to apixaban or placebo. A variable assignment sequence within blocks was generated by a centralized computer algorithm. The randomization schedule was prepared by the Investigational Drug Services at Brigham and Women's Hospital, which had no contact with the patients. All the investigators and patients were unaware of trial-group assignments. Apixaban and matching placebo were supplied by the industry sponsor. The Investigational Drug Services packaged and shipped the apixaban and placebo to the trial patients. The intended duration of administration was 12 months.

FOLLOW-UP

Virtual trial visits were conducted every 3 months in addition to an in-person visit at 12 months. At each visit, patients were assessed for medication adherence, serious adverse events, adverse drug reactions, and trial outcomes.

OUTCOME MEASURES

The primary efficacy outcome was the first symptomatic, recurrent, objectively confirmed VTE — which was defined as a composite of deep-vein thrombosis, pulmonary embolism, or both — during the 12 months after randomization, as measured in a time-to-event analysis. A secondary efficacy outcome was the first occurrence of a composite of death from cardiovascular causes, nonfatal myocardial infarction, stroke or transient ischemic attack (TIA), systemic embolism, major adverse limb event, or coronary or peripheral ischemia resulting in revascularization at 12 months. Additional exploratory composite outcomes were the first symptomatic recurrent VTE or any components of the secondary efficacy outcome and a composite of fatal or nonfatal symptomatic recurrent VTE, myocardial infarction, stroke or TIA, or systemic embolism (thrombotic outcome).

The primary safety end point was the first occurrence of major bleeding according to the criteria of the International Society on Thrombosis and Hemostasis.[19,20] A secondary safety outcome was the first occurrence of clinically relevant nonmajor bleeding.[21]

All major end points were adjudicated according to standard definitions, which are detailed in the Supplementary Appendix. An independent clinical-events committee adjudicated all end points in a blinded manner. Medication adherence was assessed according to a pill count every 3 months, along with a drug diary that was provided to patients.

STATISTICAL ANALYSIS

To determine the sample size, we estimated an incidence of 6% for symptomatic recurrent VTE in the placebo group and a 75% lower risk of this outome in the apixaban group at a two-sided alpha level of 0.05.[5,7,22] To provide 80% power to detect a significant between-group difference in the primary efficacy outcome, we estimated that each group would require 279 patients. Assuming a dropout incidence of 7% owing to loss of follow-up or postrandomization exclusion, we enrolled a total of 600 patients (300 per group).

We used descriptive statistics to describe the patients' demographic and baseline clinical characteristics, with a standardized difference of more than 10% considered to indicate clinical relevance.[23] The standardized difference between groups was calculated as the difference in means or percentages divided by the pooled standard deviation, with the type of calculation dependent on the type of variable. The first occurrence of symptomatic recurrent VTE (the primary efficacy outcome) up to and including day 365 in the intention-to-treat population was determined according to a time-to-event analysis. Supplemental analyses were conducted in a modified intention-to-treat cohort that included all the patients who had undergone randomization and received at least one dose of apixaban or placebo and in a per-protocol cohort of patients who had received at least 80% of doses of apixaban or placebo. An additional analysis was performed in the per-protocol cohort that included censoring for major or clinically relevant nonmajor bleeding. For the primary and secondary safety outcomes, analyses were performed in the safety population, which consisted of patients who had received at least

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one dose of apixaban or placebo. This analysis included the time interval from the administration of the first dose to 96 hours after the administration of the last dose.

Survival analysis was performed to compare time-to-event outcomes between groups. Cumulative incidence functions were used to estimate the percentage of patients in whom the outcome of interest was determined over time, after accounting for the competing risk of death through cause-specific hazard models.[24] A two-sided log-rank test was used to assess differences in survival distributions between groups. Hazard ratios with corresponding two-sided 95% confidence intervals were calculated. The proportional-hazards assumption was tested[25] and was not violated in any analyses. No data were found to be missing in these analyses.

Prespecified subgroup analyses were conducted on the basis of an index event of pulmonary embolism, an age 65 years or older, sex, chronic kidney disease (defined as an estimated glomerular filtration rate of <60 ml per minute per 1.72 m^2 of body-surface area), atherosclerotic cardiovascular disease, baseline aspirin use, and number of provoking factors (≥2 vs. <2). Post hoc subgroup analyses were conducted on the basis of a history of previous VTE and the presence of provoking factors of major surgery, trauma with fracture, and immobility of at least 3 days ("transient major"), as compared with those with all other provoking factors ("transient minor"). Risk differences were also calculated for subgroup analyses. Analyses were conducted with the use of R software, version 4.4.0 (R Foundation for Statistical Computing), by the trial statistician according to the statistical analysis plan, available in the protocol.

RESULTS

PATIENTS

From November 2020 through November 2023, a total of 1195 patients underwent screening (Fig. S1 in the Supplementary Appendix). Of these patients, 600 met the eligibility criteria, underwent randomization, and were included in the intention-to-treat analysis. Follow-up at 365 days after randomization was obtained in 100% of the patients.

Among the 600 patients who had undergone randomization, the mean age was 59.5 years, 57% were female, and 19.2% were of non-White race (Table 1 and Table S1). Demographic and baseline clinical characteristics were generally well balanced in the two groups.

The most common provoking factors were surgery (33.5%), immobility (31.3%), trauma (19.2%), and acute medical illness (18.3%) (Table 2 and Table S2). The most common enduring risk factors were chronic inflammatory or autoimmune disorder (52.2%), a body-mass index of 30 or more (48.2%), atherosclerotic cardiovascular disease (29.3%), and chronic lung disease (22.3%). Of a maximum of 365 days, the median number of days for adherence to the assigned dose of apixaban or placebo was 350 in the apixaban group and 339 in the placebo group (Table S3).

EFFICACY OUTCOMES

In the intention-to-treat population, symptomatic recurrent VTE (the primary efficacy outcome), which was evaluated by day 365 after randomization, occurred in 4 patients (1.3%) in the apixaban group and in 30 patients (10.0%) in the placebo group (hazard ratio, 0.13; 95% confidence interval [CI], 0.04 to 0.36; P<0.001) (Table 3 and Fig. 1A). A secondary composite outcome of cardiovascular death, nonfatal myocardial infarction, stroke or TIA, systemic embolism, major adverse limb event, or coronary or peripheral ischemia leading to revascularization was rare and occurred with similar frequency in the apixaban group and the placebo group (0.7% and 1.0%, respectively) for a hazard ratio of 0.67 (95% CI, 0.11 to 3.98) (Fig. S2).

SAFETY OUTCOMES

Major bleeding (the primary safety outcome) occurred in 1 patient in the apixaban group and none in the placebo group (0.3% vs. 0%; P>0.999) (Table 3 and Fig. 1B). The major bleeding episode was a 3-mm parafalcine subdural hematoma after a fall from a horse that resulted in neither hospitalization nor discontinuation of apixaban after neurosurgical consultation; there were no further sequelae, with uneventful on-treatment follow-up. No fatal bleeding was observed in either group.

Clinically relevant nonmajor bleeding was observed in 14 of 294 patients (4.8%) in the apixaban group and in 5 of 294 patients (1.7%) in the placebo group (hazard ratio, 2.68; 95% CI, 0.96 to 7.43; P=0.06) (Table 3, Fig. 1C, and Tables S4 and

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Table 1. Demographic and Clinical Characteristics of the Patients at Baseline.*

Characteristic	All Patients (N=600)	Apixaban (N=300)	Placebo (N=300)	Standardized Difference†
				%
Age — yr	59.5±15.2	59.1±15.2	59.9±15.2	5.8
Female sex — no. (%)	342 (57.0)	176 (58.7)	166 (55.3)	6.7
Race — no. (%)‡				
White	485 (80.8)	246 (82.0)	239 (79.7)	5.9
Black	70 (11.7)	33 (11.0)	37 (12.3)	4.2
Asian	5 (0.8)	4 (1.3)	1 (0.3)	11.0
Other	18 (3.0)	9 (3.0)	9 (3.0)	0
Unknown	27 (4.5)	9 (3.0)	18 (6.0)	14.5
Hispanic or Latino ethnic group — no. (%)‡	55 (9.2)	26 (8.7)	29 (9.7)	3.5
BMI§	30.6±6.7	30.7±6.8	30.4±6.6	3.5
Current smoker — no. (%)	32 (5.3)	15 (5.0)	17 (5.7)	3.0
Coexisting conditions — no. (%)¶				
Previous VTE‖	125 (20.8)	67 (22.3)	58 (19.3)	7.4
Diabetes	71 (11.8)	40 (13.3)	31 (10.3)	9.3
Hypertension	296 (49.3)	146 (48.7)	150 (50.0)	2.7
Coronary artery disease	104 (17.3)	50 (16.7)	54 (18.0)	3.5
Peripheral artery disease	23 (3.8)	12 (4.0)	11 (3.7)	1.7
Stroke or TIA	31 (5.2)	17 (5.7)	14 (4.7)	4.5
Heart failure	15 (2.5)	10 (3.3)	5 (1.7)	10.7
Dyslipidemia	317 (52.8)	164 (54.7)	153 (51.0)	7.4
Carotid occlusive disease	13 (2.2)	6 (2.0)	7 (2.3)	2.3
Family history of VTE — no. (%)	154 (25.7)	78 (26.0)	76 (25.3)	1.5
Use of aspirin at ≤81 mg/day — no. (%)				
At baseline	125 (20.8)	55 (18.3)	70 (23.3)	12.3
During follow-up period	119 (19.8)	54 (18.0)	65 (21.7)	9.2

* Plus–minus values are means ±SD. TIA denotes transient ischemic attack and VTE venous thromboembolism.
† The standardized difference between groups was calculated as the difference in means or percentages divided by the pooled standard deviation, with the type of calculation dependent on the type of variable.
‡ Race and ethnic group were reported by the patients.
§ The body-mass index (BMI) is the weight in kilograms divided by the square of the height in meters.
¶ None of the patients in either trial group had active cancer.
‖ A previous provoked VTE refers to an event that occurred before the occurrence of the trial eligibility event.

S5). The most common types of clinically relevant nonmajor bleeding were vaginal (1.7%), hematuria (1.4%), and rectal (1.0%) among patients in the apixaban group and vaginal (1.0%) among those in the placebo group. Death occurred in 1 patient from progressive dementia in the apixaban group and in 3 patients from newly diagnosed lung cancer, end-stage lung disease, and trauma in the placebo group.

ADVERSE EVENTS

Nonhemorrhagic, nonfatal adverse events occurred in 6 patients (2.0%) in both the apixaban and placebo groups. Adverse events were associated with

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Table 2. Characteristics of Venous Thromboembolic (VTE) Events and Risk Factors (Intention-to-Treat Population).*

Characteristic	Total (N=600)	Apixaban (N=300)	Placebo (N=300)	Standardized Difference
				%
VTE diagnosis for eligibility — no. (%)				
Only deep-vein thrombosis	288 (48.0)	142 (47.3)	146 (48.7)	2.7
Isolated calf deep-vein thrombosis	119 (19.8)	48 (16.0)	71 (23.7)	19.3
Only pulmonary embolism	140 (23.3)	77 (25.7)	63 (21.0)	11.1
Right ventricular dysfunction	80 (13.3)	38 (12.7)	42 (14.0)	3.9
Both pulmonary embolism and deep-vein thrombosis	172 (28.7)	81 (27.0)	91 (30.3)	7.4
Provoking factors for VTE — no. (%)†				
Acute medical illness‡	110 (18.3)	56 (18.7)	54 (18.0)	1.7
Surgery	201 (33.5)	102 (34.0)	99 (33.0)	2.1
Trauma	115 (19.2)	62 (20.7)	53 (17.7)	7.6
Pregnancy	11 (1.8)	7 (2.3)	4 (1.3)	7.5
Infection	99 (16.5)	45 (15.0)	54 (18.0)	8.1
Hormonal contraceptive or replacement therapy	69 (11.5)	42 (14.0)	27 (9.0)	15.7
Hospitalization ≤3 mo before the VTE event	56 (9.3)	25 (8.3)	31 (10.3)	6.9
Immobility	188 (31.3)	81 (27.0)	107 (35.7)	18.8
Blood transfusion§	2 (0.3)	1 (0.3)	1 (0.3)	0
Coronavirus disease 2019§	49 (8.2)	23 (7.7)	26 (8.7)	3.7
Long-haul travel§	100 (16.7)	46 (15.3)	54 (18.0)	7.2
Other factor¶	53 (8.8)	26 (8.7)	27 (9.0)	1.2
Enduring risk factors for VTE — no. (%)†				
Persistent immobility	39 (6.5)	15 (5.0)	24 (8.0)	12.2
Obesity: BMI ≥30	289 (48.2)	141 (47.0)	148 (49.3)	4.7
Heart failure	15 (2.5)	10 (3.3)	5 (1.7)	10.7
Chronic lung disease	134 (22.3)	78 (26.0)	56 (18.7)	17.7
Chronic inflammatory or autoimmune disorder	313 (52.2)	152 (50.7)	161 (53.7)	6.0
Atherosclerotic cardiovascular disease	176 (29.3)	86 (28.7)	90 (30.0)	2.9
Chronic kidney disease	64 (10.7)	32 (10.7)	32 (10.7)	0
Chronic liver disease	23 (3.8)	10 (3.3)	13 (4.3)	5.2
Median duration of anticoagulant treatment before randomization — mo (IQR)	6.6 (3.7–35.3)	7.6 (3.9–39.6)	5.9 (3.5–32.5)	6.9

* IQR denotes interquartile range.
† Patients could have more than one provoking or enduring risk factor for VTE.
‡ This category includes exacerbation of asthma, Crohn's disease, heart failure, chronic obstructive pulmonary disease, or other inflammatory condition.
§ The listed provoking factor occurred within 30 days before the index VTE event.
¶ Details regarding other provoking factors are provided in Table S2 in the Supplementary Appendix.

permanent discontinuation of apixaban in 2 patients (0.7%) and in no patients in the placebo group (Tables S4, S5, and S6). All adverse events were classified by the data and safety monitoring board as nonserious and either unrelated or unlikely to be related to the trial-group assignment.

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Table 3. Prespecified Efficacy and Safety Outcomes (Intention-to-Treat Population).*

Outcome	Apixaban	Placebo	Hazard Ratio (95% CI)	P Value
	no./total no. (%)			
Efficacy outcome				
Symptomatic recurrent VTE: primary efficacy outcome	4/300 (1.3)	30/300 (10.0)	0.13 (0.04–0.36)	<0.001
Pulmonary embolism				
Any	0	11/300 (3.7)		
Isolated subsegmental	0	1/300 (0.3)		
Deep-vein thrombosis				
Any	4/300 (1.3)	23/300 (7.7)		
Isolated calf	2/300 (0.7)	10/300 (3.3)		
Secondary efficacy composite outcome	2/300 (0.7)	3/300 (1.0)†	0.67 (0.11–3.98)‡	
Nonfatal myocardial infarction	1/300 (0.3)	0		
Stroke, TIA, or systemic embolism§	1/300 (0.3)	2/300 (0.7)		
Coronary or peripheral ischemia resulting in revascularization	0	2/300 (0.7)		
Other outcome				
Additional efficacy composite outcome¶	6/300 (2.0)	32/300 (10.7)	0.18 (0.07–0.43)‡	
Composite thrombotic outcome‖	6/300 (2.0)	31/300 (10.3)	0.19 (0.08–0.44)‡	
Death from any cause	1/300 (0.3)	3/300 (1.0)		
Safety outcome				
Major bleeding: primary safety outcome**	1/294 (0.3)	0	NE	>0.999
Clinically relevant nonmajor bleeding: secondary safety outcome**	14/294 (4.8)	5/294 (1.7)	2.68 (0.96–7.43)	0.06

* The primary and secondary efficacy outcomes were evaluated in the intention-to-treat population of 600 patients. The safety outcomes were evaluated in the safety population of 588 patients. No patients died from cardiovascular causes or had a major limb event (both secondary efficacy outcomes). NE denotes not evaluable.

† One patient who had a stroke also underwent coronary revascularization.

‡ In this category, the widths of the confidence intervals have not been adjusted for multiplicity, so the intervals may not be used in place of hypothesis testing.

§ All three events in this category were ischemic strokes.

¶ The additional efficacy composite outcome includes all the components of the secondary efficacy outcome, as well as symptomatic recurrent VTE. The first occurrence of any of these events was included in the analysis of this outcome.

‖ The composite thrombotic outcome includes symptomatic recurrent VTE, fatal or nonfatal myocardial infarction, stroke or TIA, or systemic embolism. Death from any cause was not a component of the composite efficacy outcomes.

** Major and clinically relevant nonmajor bleeding were defined according to the criteria of the International Society on Thrombosis and Hemostasis. The single major bleeding episode was a 3-mm parafalcine subdural hematoma caused by a fall from a horse that resulted in neither hospitalization nor discontinuation of apixaban after neurosurgical consultation; no further sequelae were reported.

SUBGROUP AND SUPPLEMENTARY ANALYSES

Results with respect to the primary efficacy outcome were generally consistent across subgroups (Figs. S3 and S4). Results were also consistent across different analysis populations (Tables S7 through S12). With respect to major or clinically relevant nonmajor bleeding, results were similar in subgroups, including the use of low-dose aspirin (Figs. S5 and S6).

DISCUSSION

In this trial at 1 year, patients with provoked VTE and at least one enduring risk factor for recurrent

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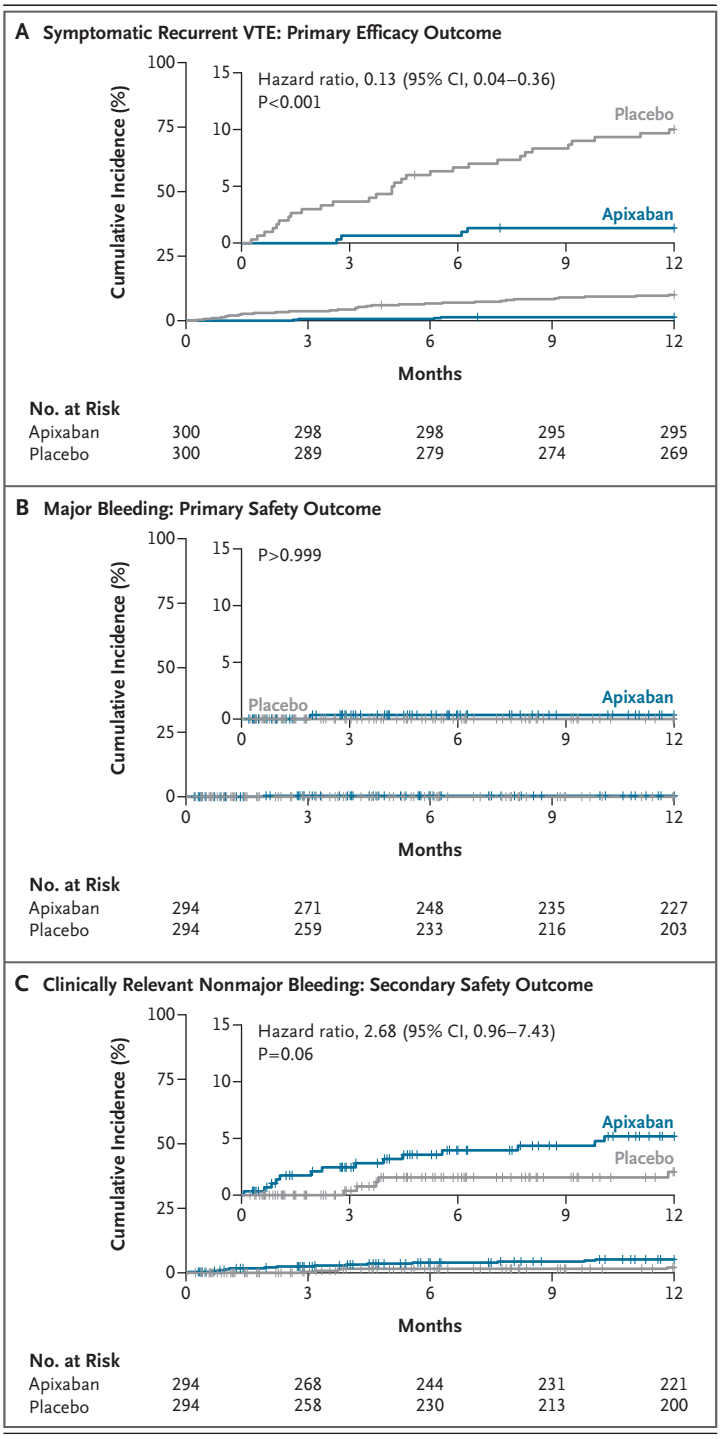
Figure 1. Symptomatic Recurrent Venous Thromboembolism (VTE), Major Bleeding, and Clinically Relevant Nonmajor Bleeding.

Shown are Kaplan–Meier curves for the first event of symptomatic recurrent VTE during a 12-month period (the primary efficacy outcome) (Panel A), as well as the first episodes of major bleeding (the primary safety outcome) (Panel B) and clinically relevant nonmajor bleeding (secondary safety outcome) (Panel C) during the period between administration of the first dose of apixaban or placebo and 96 hours after administration of the last dose. In each panel, the inset shows the same data on an enlarged y axis.



events who received apixaban (at a dose of 2.5 mg orally twice daily) had a significantly lower risk of symptomatic recurrent VTE — the primary efficacy outcome — than those who received placebo. Overall, patients in the apixaban group had few adverse events and a high degree of adherence to the trial regimen. The benefits for the prevention of recurrent VTE with extended-duration apixaban were observed with a low risk of major bleeding. The frequency of clinically relevant nonmajor bleeding appeared to be greater in patients receiving apixaban.

In our trial, the 12-month risk of symptomatic recurrent VTE was 10% in the placebo group. In the context of other observational studies and randomized trials, these data suggest that a dichotomous approach of provoked versus unprovoked VTE is insufficient to determine the duration of anticoagulation.[7,9,26] Furthermore, these data suggest that the presence of enduring risk factors may identify patients with a risk of recurrent VTE that approximates that of high-risk patients (>8%).[11] A systematic review of the risk of recurrence after discontinuation of anticoagulation in patients with provoked VTE estimated a cumulative incidence of 5.8% per patient-year (95% CI, 3.2 to 8.3%).[9] The higher observed event rate in our trial may be related to a combination of the presence of enduring risk factors and chance, as reflected by the lower boundary of the 95% confidence interval for the cumulative annual incidence of recurrent VTE events in the placebo group (6.6%), which more closely fits previous observations. Current guidelines do not routinely recommend extended-duration anticoagulation for patients with provoked VTE.[4,6,8,11] However, recognition of the detrimental role of enduring risk factors (e.g., chronic inflammatory conditions, obesity, and chronic lung disease) may facilitate more precise risk assessment.[18] Our results highlight a population with provoked VTE and enduring risk factors in which the risk of

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recurrence after the discontinuation of anticoagulation is high enough to warrant consideration of continued anticoagulation.

Extended-duration apixaban decreased the risk of symptomatic recurrent VTE by 87% in patients who had provoked VTE and at least one enduring risk factor for recurrence. These findings mirror data from trials of extended-duration secondary prevention in patients with unprovoked VTE,[2] as well as in a mixed population of patients with both provoked and unprovoked events.[7,27,28] In contrast to previous trials, HI-PRO exclusively focused on an underrepresented population in the literature: patients with provoked VTE and at least one enduring risk factor for recurrence. Of note, in a post hoc subgroup analysis, we observed that the treatment effect for reducing the risk of recurrent VTE was consistent in patients with both major and minor transient provoking factors.

In this trial, we found that apixaban (at a dose of 2.5 mg twice daily) was efficacious for the prevention of symptomatic recurrent VTE with a low risk of major bleeding, results that are similar to those of other trials of low-intensity anticoagulation with direct oral drugs.[2,7] Also consistent with the findings in these trials, we observed a higher incidence of clinically relevant nonmajor bleeding in the apixaban group than in the placebo group (4.8% vs. 1.7%), a result that mirrored data from other trials of low-intensity anticoagulation for long-term secondary prevention (2.0 to 8.6%).[2,7,27] Clinically relevant nonmajor bleeding events are associated with increased complications, decreased quality of life, and increased health care use and costs.[29] The hazard of clinically relevant nonmajor bleeding highlights the need to assess bleeding risk in addition to recurrence risk when engaging in shared decision making focused on extended-duration anticoagulation.[18,30] Since none of the recurrent VTE events were fatal in our trial, the risk–benefit trade-off with clinically relevant nonmajor bleeding may be nuanced, depending on patient preferences and previous experiences. Of note, nearly 20% of the trial patients (without major between-group difference) continued to receive aspirin throughout the follow-up period. Discontinuation of aspirin in patients receiving anticoagulation who are in stable condition has been shown to reduce bleeding in other clinical scenarios.[31,32]

Additional research involving patients with provoked VTE and enduring risk is needed to further inform guideline-writing committees and clinical practice. Studies that have used advanced risk-stratification tools — such as the VTE-PREDICT risk score to estimate the effect of extended anticoagulant treatment,[18,33] polygenic risk scores,[34,35] and artificial intelligence[36] — may more precisely identify populations in which the risk–benefit ratio can be maximized. Cost-effectiveness and patient-informed outcome analyses are warranted since factors regarding extended-duration anticoagulation — health care use and cost, along with quality-of-life issues — are not always apparent.[37] In the meantime, the results of HI-PRO suggest that clinical assessment of enduring risk factors can identify patients who may benefit from extended-duration anticoagulation and that low-intensity apixaban in such patients is feasible and associated with a high level of medication adherence.

Our trial should be interpreted within the context of its limitations. Because of its single-center design, the results may not be readily generalizable to other health care systems. Although we enrolled patients with VTE and a broad range of major and minor transient provoking factors and enduring risk factors representative of those presenting in clinical practice, the trial was neither designed nor powered to identify differences in outcomes according to specific risk contributors. The racial composition of the overall trial population was similar to those in large observational studies, and 20% of the patients were non-White, which improves on numbers reported in previous studies of secondary prevention of VTE. However, greater racial and ethnic diversity in trials of therapies for thrombotic diseases is needed (Table S1).[38] An important difference in clinically relevant nonmajor bleeding may have been missed since HI-PRO was not powered for the analysis of that end point. Finally, the ability to discern differences in outcomes among subgroups — including those with specific transient provoking circumstances, enduring risk factors, or their combination — was limited by the sample size.

Oral apixaban (at a dose of 2.5 mg twice

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daily) was effective for the prevention of symptomatic recurrent VTE in patients with provoked VTE and at least one enduring risk factor for recurrence. The reduction in the risk of symptomatic recurrent VTE in patients receiving apixaban was associated with a low risk of major bleeding.

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